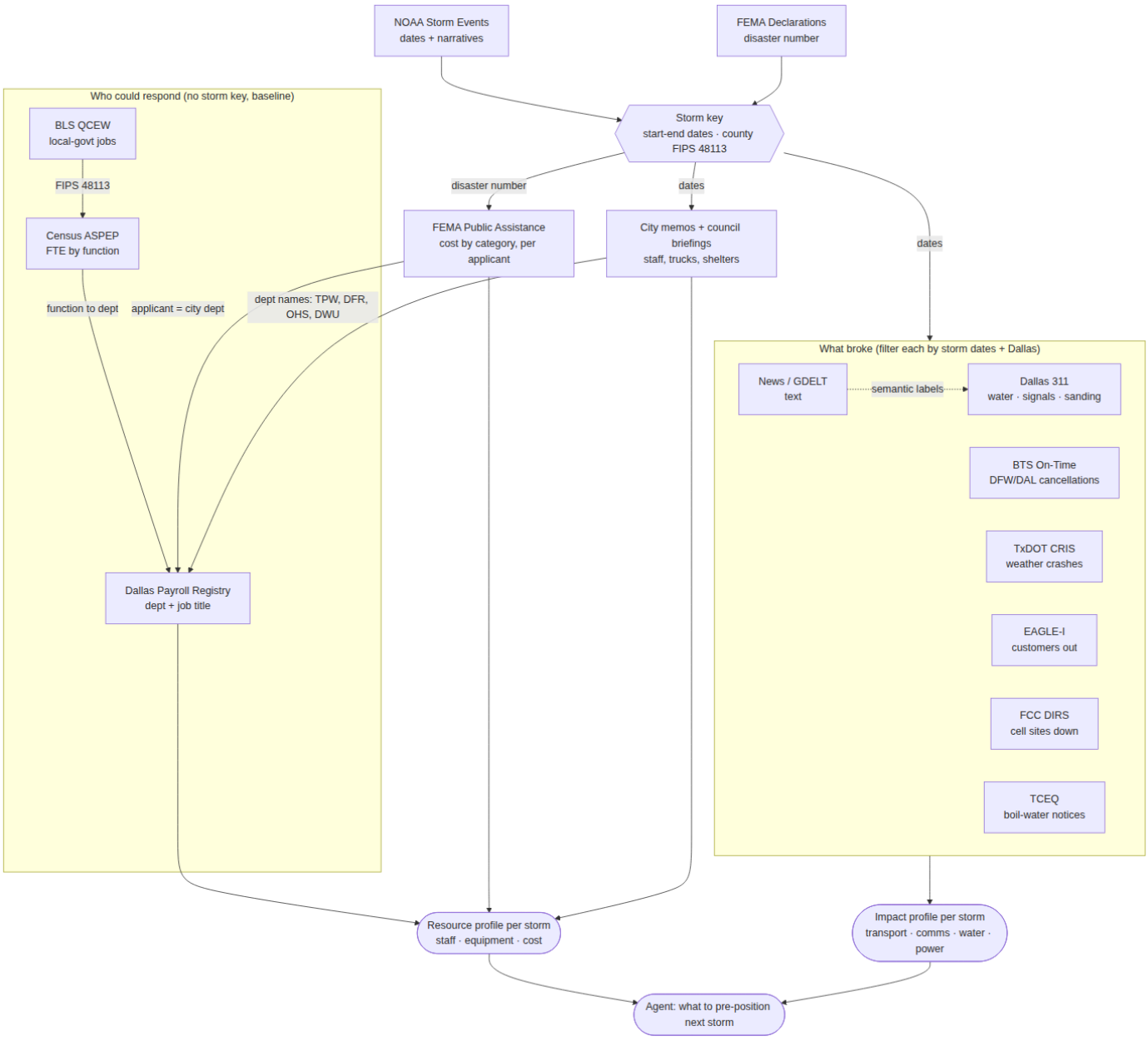
Dallas County Storm Data Sources

Data sources for a county-level storm-readiness agent, focused on **Dallas County, TX (FIPS 48113)**. This doc lists sources only. The counts are there to show each source really has Dallas coverage; they are not the data itself. Checked 2026-10-02.

Why Dallas, and which storms

- **Hurricanes don't hit Dallas directly. They reach it in two ways:**
    1. **Evacuee sheltering.** Dallas County appears on 9 FEMA hurricane or tropical-storm declarations: Tropical Storm Charley 1998, Katrina 2005, Rita 2005 (×2), Dean 2007, Gustav 2008, Ike 2008, Harvey 2017, Marco/Laura 2020. Most are emergency (EM) declarations for sheltering evacuees, not local damage. Katrina brought about 26,000 evacuees to North Texas shelters; Harvey had 3,500+ people at the Dallas convention center.
    2. **Remnant rainfall and flooding.** NOAA records 5 Dallas events tied to Tropical Storm Hermine (2010), 2 to Lorena's remnants (2025), and 13 flood events in August 2022 driven by tropical moisture (about 1.35B in recorded property damage).
- **Winter storms do direct damage:** Uri 2021, the 2023 ice storm, January 2025, January 2026. NOAA has 80 winter-type events for Dallas from 2000 to 2026.
- **Outlook:** a strong El Niño is expected for winter 2026–27, which makes a wetter North Texas likely. Ice also needs arctic air at the same time, so this raises the odds; it is not a forecast.

How the sources connect



The join key is the storm window. NOAA event dates plus the FEMA disaster number define each storm. Every impact source is then filtered by those dates and by Dallas (FIPS 48113, city limits, or the DFW and Love Field airports). The staffing sources don't have a storm key. They describe the baseline workforce, and they link to storm-time resource use through department names: TPW = Transportation & Public Works, DFR = Dallas Fire-Rescue, OHS = Office of Homeless Solutions, DWU = Dallas Water Utilities. Those names appear in the city memos, FEMA applicants, and 311 request types.

1. Storm identification

Source	What it has	Access	Dallas coverage (checked)
NOAA Storm Events Database	Each event's type, start and end, deaths, injuries, damage, and **free-text episode and event narratives**	Bulk CSV: `ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/`, filter `STATE=TEXAS`, `CZ_NAME=DALLAS`	1,410 events (2000 to Jun 2026); 1,368 with narratives
OpenFEMA Disaster Declarations Summaries	Disaster number, incident type, which programs were declared (PA, IA) for each county	API: `fema.gov/api/open/v2/DisasterDeclarationsSummaries`, filter `fipsStateCode='48'`, `fipsCountyCode='113'`	28 declarations (1966–2024), 9 of them hurricane or tropical storm

2. Workforce: who could respond (baseline)

Source	What it has	Access	Notes
City of Dallas Payroll Registry	City-wide **totals** every two weeks: civilian vs uniformed employee count and pay. No department or job title	Dallas OpenData `qrgv-mijx` (Socrata API)	134 rows, last updated 2019. Only useful as a headcount baseline (e.g. about 7,800 civilian, 5,100 uniformed in Oct 2019)
City of Dallas Operating Budget	Budget by appropriation, service, and object group (personnel, supplies, capital)	Dallas OpenData `e2fs-y4nb`	1,498 rows, FY2026. Personnel **spending** by service, not headcount. Headcount by department is only in the annual budget book (PDF)
Census Annual Survey of Public Employment & Payroll (ASPEP), individual-unit file	Full-time-equivalent staff and payroll by **function** (police, fire, highways, water, hospitals, corrections) for each government	census.gov ASPEP downloads	Covers Dallas County, the cities in it, and special districts. Only source that gives function for every local government
BLS QCEW	Local-government employment and wages by county and industry, quarterly	bls.gov/cew	County totals; not checked this session

3. Resources used in past storms

Source	What it has	Access	Dallas coverage (checked)
City memo "January 2025 Winter Storm by the Numbers"	Staff by department, trucks, lane-miles sanded, shelters and occupancy, calls by type	dallascityhall.com, City Manager memos FY24-25	175 TPW staff on two 12-hour shifts, 50 trucks, 4,000 lane-miles; 200+ field staff, 27 in the EOC, 180+ remote; 4 shelters averaging about 950 people a day

Source	What it has	Access	Dallas coverage (checked)
Office of Emergency Management "Winter Weather Operations" council briefing (2025-11-05)	Shelter history 2019–2025, cost per activation, departments involved, comparison with Houston, Austin, Fort Worth, San Antonio	cityofdallas.legistar.com	Peak 1,314 guests; 4,167 stays in January 2025; FY24 cost 2.9M against 0.9M budgeted
Dallas City Council Legistar archive	Every briefing and after-action presentation, searchable by date	cityofdallas.legistar.com	Main source for Uri, 2023, and 2026 staffing (needs a scraper)
Garland after-action (January 2026)	Calls, sand, staff hours for a city inside Dallas County	Garland council video and agenda	2,915 police calls, 591 fire calls, 200 t sand
OpenFEMA Public Assistance Funded Projects Details	Each funded project's applicant, damage category (A debris … G recreation; B = emergency protective measures), and amount	API: `fema.gov/api/open/v2/PublicAssistanceFundedProjectsDetails`, filter `countyCode='113'`	516 projects. Katrina (EM-3216): 82 projects, about 51.8M, almost all sheltering and interim housing. Uri (DR-4586): 33 projects, about 4.2M
City of Dallas recovery-grant (CDBG-DR) plan, Winter Storm Uri	Unmet needs and allocation after Uri	dallascityhall.com, CDBG-DR page	40M+ allocated

FEMA caveat: the PA `county` field is the **applicant's** county, not where the damage was. The Ike and Harvey records under Dallas are mostly a hospital system headquartered in Dallas County repairing hospitals on the coast. Filter by applicant before treating these as local.

4. Impacts: what broke, by sector

Sector	Source	What it has	Access	Dallas coverage (checked)
Cross-sector	**Dallas 311 Service Requests (2000 to present)**	Request type, department, created and closed time, location	Dallas OpenData `4n3h-rfqm` (Socrata API, updated hourly)	5.36M requests in total. Storm windows: Uri 3,527; Feb 2023 ice 4,501; Jan 2025 2,356; Jan 2026 3,800. Weather-specific types include "Snow/Ice Sanding – Hazardous", "Traffic Signal – Flashing", "Water/Wastewater", "Inclement Weather Shelter Transportation"

Sector	Source	What it has	Access	Dallas coverage (checked)
Transportation (air)	**BTS Reporting Carrier On-Time Performance**	Every flight's cancellation flag and cause code (B = weather), for DFW and DAL	transtats.bts.gov, monthly zip	Not pulled. News: about 75% of DFW flights cancelled in Feb 2023; about 1,500 at DFW and Love Field in Jan 2026
Transportation (roads)	**TxDOT CRIS crash records**	Crashes by county with weather and surface condition (ice)	cris.dot.state.tx.us public query	Not pulled
Power	**EAGLE-I (ORNL)**	Customers without power by county, every 15 minutes, 2014–2025	figshare doi 10.6084/m9.figshare.24237376; OSTI	Already in the project's data contract
Communications	**FCC DIRS reports**	Cell sites out of service by county; status of 911 centers, cable, and wireline during an activation	fcc.gov DIRS daily reports (PDF/TXT)	Activated for Harvey 2017 and Winter Storm Fern (Jan 2026); not confirmed for Uri
Water	**TCEQ boil-water notices / Drinking Water Watch**	Notices by water system and date	tceq.texas.gov	Not pulled; 311 water types work as a proxy
Narrative (for the news analysis)	**NOAA narratives + local news + GDELT**	Free text: what failed and why	NOAA bulk CSV; GDELT API; KERA, Dallas Morning News, WFAA	Start with the NOAA narratives (1,368 Dallas texts, already geocoded)

5. Hurricane evacuee sheltering (Dallas as a host city)

For Dallas the hurricane risk is **receiving** coastal evacuees, not evacuating its own residents. The questions are: how many arrive, where they stay, and which local staff and facilities get used.

Question	Source	What it has	Access	Dallas coverage
Sheltering cost per storm	**OpenFEMA Public Assistance, category B** (section 3)	Shelter and interim-housing projects per applicant	API, filter `countyCode='113'`	Katrina EM-3216: 82 projects, about 51.8M. Also Rita, Dean, Gustav, Ike, Harvey, Laura
Shelter capacity (baseline)	**National Shelter System Facilities** (FEMA and Red Cross)	Each shelter's location, capacity, ADA access, pets	data.gov `national-shelter-system-facilities`; mirrors at the Data Rescue Project and FEMA on ArcGIS (the government site HIFLD Open has been shut down)	All designated Dallas shelters
Shelters open now	**National Shelter System, open shelters**	Open shelters and occupancy	AmeriGEOSS / FEMA feed	Live only, no history
State hub plan	**Texas Mass Care annex (ESF-6), 2020**	How the shelter hub system, reception centers, and shelter types work	Texas Division of Emergency Management (TDEM) website, PDF	Doesn't list hub cities or capacity; needs a request to TDEM
Load on hospitals	**CDC: medical surge in DFW during Harvey**	Emergency-department visits at 55 DFW hospitals (North Texas syndromic surveillance); evacuees' home counties	stacks.cdc.gov/view/cdc/86104	About 16,500 excess visits; visits from patients in disaster-declared counties +127%
Shelter operations	**Harvey mass sheltering in Dallas County** (PMC6255530)	Disease surveillance at the Dallas mega-shelter	PMC	3,500+ registered guests
Shelter shortfall method	**Harvey shelter deficits for socially vulnerable Texans** (PMC6537281)	Shelter shortfall by county and population group	PMC	Method we could reuse
City staffing and cost	**Dallas council archive (Legistar) + 311**	Mega-shelter briefings; 311 requests for the storm window	Legistar; Dallas OpenData `4n3h-rfqm`	Harvey window: 16,877 requests in 311, mostly routine types, so filter for shelter-related ones

Question	Source	What it has	Access	Dallas coverage
Where evacuees go	**Texas Hurricane Evacuation Studies** (USACE, FEMA, Texas A&M Coastal Bend)	Behavioral surveys: share evacuating inland, destinations	texasatlas.arch.tamu.edu	No public DFW-specific share found
Where evacuees go	**Smartphone location data to predict evacuee destinations** (Springer, 2023)	Machine-learning method	Paper	Underlying data usually not public
Where evacuees ended up	**Katrina displacement counts** (Census and surveys)	Where evacuees ended up	Census	North Texas about 26,000 in shelters

Gap: no public dataset counts daily evacuee arrivals in Dallas. Best proxy: ER visits from disaster-declared counties + FEMA category B cost + peak shelter counts from council briefings.

Gaps

1. **Staff deployed per storm** only exists as numbers for January 2025. Other storms need the Legistar briefings scraped and pulled into a table by hand.
2. **No per-employee roster is open data**, for the city or the county. Headcount by department means the budget book PDF, or ASPEP at the function level.
3. **The news analysis has a small sample:** about 4–5 major storms. Use the structured counts (311, BTS, EAGLE-I, DIRS) to measure impact, and the text to explain why.
4. **Hurricane relevance:** for Dallas the hurricane problem is mostly *sheltering evacuees* (section 5), not wind or surge damage. Pick the agent's framing before choosing which sources matter most.